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1. Groups

1.1. Notation

1. $\mathbb{N} = \{1, 2, 3, \dots\}$
2. $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
3. $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{N}\}$
4. $\mathbb{R}$ = the set of real numbers
5. $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$
6. For $n \in \mathbb{N}$, $\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$, where congruence class $[r] = \{z \in \mathbb{Z} : z \equiv r \pmod{n}\}$ for $0 \leq r \leq n-1$.
7. For $n \in \mathbb{N}$, an $n \times n$ matrix over $\mathbb{R}$ (or $\mathbb{Q}$ or $\mathbb{C}$) is an $n \times n$ array
$$A = [a_{i,j}] = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix}$$
8. $M_n(\mathbb{R})$ = the set of $n \times n$ matrices over $\mathbb{R}$ with the usual addition and multiplication

1.2. Groups

Definition 1.2.1

G set, $\star$ operation on $G \times G$. $(G, \star)$ is a group if it satisfies the following:

1. $\star$ is associative (if $a, b, c \in G$, then $a \star (b \star c) = (a \star b) \star c$)
2. There exists an (identity) element $e \in G$ such that $a \star e = a = e \star a$ for all $a \in G$
3. For each $a \in G$, there exists an inverse $a^{-1} \in G$ such that $a \star a^{-1} = e = a^{-1} \star a$

Definition 1.2.2

A group G is said to be abelian if it is commutative.

Exercise 1.2.3

Prove that in the definition of a group, it suffices to assume only that $e \star a = a$ in (3) and $a^{-1} \star a = e$ in (4).

Proposition 1.2.4

Let G be a group as defined in the previous exercise. Then $a \star e = a$ and $a \star a^{-1} = e$.

Proof:

$$\begin{aligned}
e &= e \star e \\
&= (a^{-1} \star a) \star (a^{-1} \star a) \\
\Rightarrow a \star e \star a^{-1} &= a \star (a^{-1} \star a) \star (a^{-1} \star a) \star a^{-1} \\
\Rightarrow a \star a^{-1} &= (a \star a^{-1})^3 \\
\Rightarrow e \star (a \star a^{-1}) &= (a \star a^{-1})^2 \star (a \star a^{-1}) \\
\Rightarrow e &= (a \star a^{-1}) \star (a \star a^{-1}) \\
\Rightarrow e &= a \star a^{-1} \\
\Rightarrow a \star e &= a \star (a^{-1} \star a) \\
&= (a \star a^{-1}) \star a \\
&= e \star a
\end{aligned}$$

□

Proposition 1.2.5

Let G be a group, and let $a \in G$. Then

1. The identity of G is unique
2. The inverse of a is unique.

Proof:

1. Let e_1, e_2 be identities. Since e_2 is a right identity and e_1 is a left identity, then $e_1 = e_1 \star e_2 = e_2$.
2. Let a_1, a_2 be inverses of a . Then $a_1 = e \star a_1 = (a_2 \star a) \star a_1 = a_2 \star (a \star a_1) = a_2 \star e = a_2$.

□

Example

The sets $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$ are abelian groups, where the additive identity is 0 and the additive inverse of r is $-r$.

Note that $(\mathbb{N}, +)$ is not a group, since it has no identity.

Example

The sets $(\mathbb{Q}, \cdot)$, $(\mathbb{R}, \cdot)$ and $(\mathbb{C}, \cdot)$ are not groups as 0 has no multiplicative inverse in $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$.

Example

For set $S = (S, \cdot)$, let S^* denote the subset of S containing all elements with multiplicative inverses.

For example, $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$. Then $(\mathbb{Q}^*, \cdot)$, $(\mathbb{R}^*, \cdot)$ and $(\mathbb{C}^*, \cdot)$ are abelian groups, where the multiplicative identity is 1 and the multiplicative inverse of r is $-\frac{1}{r}$.

Exercise 1.2.6

What is $\mathbb{Z}_n^*$?

Solution:

$$\mathbb{Z}_n^* = \{[k] : \gcd(k, n) = 1\}$$

Example

The set $M_n(\mathbb{R})$ is an abelian group where the additive identity is the zero matrix and the additive inverse of A is $-A$.

Example

Consider $(M_n(\mathbb{R}), \cdot)$. The identity matrix is $I \in M_n(\mathbb{R})$.

However, since not all $M \in M_n(\mathbb{R})$ has a multiplicative inverse, $(M_n(\mathbb{R}), \cdot)$ is not a group.

Definition 1.2.7

Define the set $GL_n(\mathbb{R}) = \{M \in M_n(\mathbb{R}) : \det(M) \neq 0\}$.

Note

If $A, B \in GL_n(\mathbb{R})$, then $\det(A) \cdot \det(B) \neq 0$ and thus $AB \in GL_n(\mathbb{R})$. The associativity of $GL_n(\mathbb{R})$ inherits from $M_n(\mathbb{R})$'s associativity. Also, the identity matrix I satisfies $\det(I) = 1 \neq 0$ and thus $I \in GL_n(\mathbb{R})$. Finally, for $M \in GL_n(\mathbb{R})$, there exists $M^{-1} \in M_n(\mathbb{R})$ such that $MM^{-1} = I = M^{-1}M$. Since $\det(M^{-1}) = \frac{1}{\det(M)} \neq 0$, we have $M^{-1} \in GL_n(\mathbb{R})$. Thus $(GL_n(\mathbb{R}), \cdot)$ is a group.

Definition 1.2.8

$(GL_n(\mathbb{R}), \cdot)$ is called the general linear group of degree n over $\mathbb{R}$.

Note

If $n \geq 2$, the group $(GL_n(\mathbb{R}), \cdot)$ is not abelian.

Exercise 1.2.9

What is $(GL_1(\mathbb{R}), \cdot)$?

Solution:

$$(GL_1(\mathbb{R}), \cdot) \cong (\mathbb{R} \setminus \{0\}, \cdot)$$

Example

Let G and H be groups. Then the direct product is the set $G \times H$ with the component-wise operation defined by $(g_1, h_1) \star (g_2, h_2) = (g_1 \star_G g_2, h_1 \star_H h_2)$. One can check that $G \times H$ is a group with the identity (e_G, e_H) and the inverse of (g, h) is (g^{-1}, h^{-1}) .

Note

One can show by induction that if $G_1, G_2, \dots, G_n$ are groups, then $G_1 \times G_2 \times \dots \times G_n$ is also a group.

Notation

Given a group G and $g_1, g_2 \in G$, we often denote $g_1 \star g_2$ by $g_1 g_2$ and the identity by 1.

Also, the unique inverse of an element $g \in G$ is denoted by g^{-1} .

For $n \in \mathbb{N}$, we define $g^n = g \star g \star \dots \star g$ (n times) and $g^{-n} = (g^{-1})^n$

Finally, we denote $g^0 = 1$.

Proposition 1.2.10

Let G be a group and $g, h \in G$. We have

1. $(g^{-1})^{-1} = g$
2. $(gh)^{-1} = h^{-1}g^{-1}$
3. $g^n g^m = g^{n+m} \forall n, m \in \mathbb{Z}$
4. $(g^n)^m = g^{nm} \forall n, m \in \mathbb{Z}$

Proof:

1. Since $g^{-1}g = 1 = gg^{-1}$, then $(g^{-1})^{-1} = g$.
2. $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = g1g^{-1} = gg^{-1} = 1$ Similarly, $(h^{-1}g^{-1})(gh) = 1$.
Thus, $(gh)^{-1} = h^{-1}g^{-1}$.

□

Exercise 1.2.11

Prove (3) and (4) from the previous proposition.

Remark

In general, it is not true that if $g, h \in G$, then $(gh)^n = g^n h^n$. This is not true unless G is abelian.

Proposition 1.2.12

Let G be a group and $g, h, f \in G$. The following are true:

1. They satisfy left and right cancellation. More precisely,
 - a. if $gh = gf$, then $h = f$.
 - b. if $hg = fg$, then $h = f$.
2. Given $a, b \in G$, the equations $ax = b$ and $ya = b$ have unique solutions for $x, y \in G$.

Proof: (1.a): By left-multiplying by g^{-1} , we have

$$\begin{aligned} gh &= gf \\ \Rightarrow g^{-1}(gh) &= g^{-1}(gf) \\ \Rightarrow h &= f \end{aligned}$$

The proof of (1.b) is similar.

(2): Let $x = a^{-1}b$. Then $ax = a(a^{-1}b) = (aa^{-1})b = 1b = b$.

If u is another solution, then $au = b = ax$. By (1.a), $u = x$.

Similarly, $y = ba^{-1}$ is the unique solution of $ya = b$. □

1.3. Symmetric groups**Definition 1.3.1**

Given a non-empty set L , a permutation of L is a bijection from L to L . The set of all permutations of L is denoted by S_L .

Example

Consider the set $L = \{1, 2, 3\}$, which has the following different permutations:

$\begin{pmatrix} 123 \\ 123 \end{pmatrix}, \begin{pmatrix} 123 \\ 132 \end{pmatrix}$ (it indicates the bijection $\sigma : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ with $\sigma(1) = 1, \sigma(2) = 3, \sigma(3) = 2$).

Notation

For $n \in \mathbb{N}$, we denote $\cdot$ by $S_n = S_{\{1, 2, \dots, n\}}$ the set of all permutations of $\{1, 2, \dots, n\}$. We have seen that the order (size) of $S_3 = 3! = 6$. To consider the order of the general S_n , we note that for a permutation of S_n , there are n choices for $\sigma(1)$, $(n - 1)$ choices for $\sigma(2)$, ..., 1 choice for $\sigma(n)$.

Proposition 1.3.2

We have $|S_n| = n!$

Note

On Mobius Quiz, use “nine dots” to format a permutation.

Given $\sigma, \tau \in S_n$, we can compose them to get a third element $\sigma\tau$, where $\sigma\tau : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ given by $x \mapsto \sigma(\tau(x)) \forall x \in \{1, 2, \dots, n\}$. Since both σ, τ are bijections, so is $\sigma\tau$. Thus $\sigma\tau \in S_n$.

Example

Compute $\sigma\tau$ and $\tau\sigma$ if $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}$

We note that $\sigma\tau \neq \tau\sigma$. For any $\sigma, \tau \in S_n$, we have $\sigma\tau, \tau\sigma \in S_n$, but $\sigma\tau \neq \tau\sigma$ in general. On the other hand, for $\sigma, \tau, \mu \in S_n$, we have that $\sigma(\tau\mu) = (\sigma\tau)\mu$.

Also, the identity permutation $\varepsilon \in S_n$ is defined as

$$\varepsilon = \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix}$$

Then for any $\sigma \in S_n$, we have $\sigma\varepsilon = \sigma = \varepsilon\sigma$.

Finally, for $\sigma \in S_n$, since it is a bijection, there exists a unique bijection σ^{-1} , called the inverse permutation of σ , such that for all $x, y \in \{1, 2, \dots, n\}$, $\sigma^{-1}(x) = y$ iff $\sigma(y) = x$. It follows that $\sigma(\sigma^{-1}(x)) = x$ and $\sigma^{-1}(\sigma(y)) = y$.

So we have $\sigma\sigma^{-1} = \varepsilon = \sigma^{-1}\sigma$.

For example, find the inverse of $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 2 & 3 \end{pmatrix}$.

Note that $\sigma(1) = 4$. Thus $\sigma^{-1}(4) = 1$. Compute the rest similarly.

We have $\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix}$.

Proposition 1.3.3

S_n is a group under function composition, called the symmetric group of degree n .

Proof: Above example. □

Exercise 1.3.4

Write down all rotations and reflections that fix an equilateral triangle. Then check why it is the same as S_3 .

Solution:

$e \in S_3$ corresponds to no rotations nor reflections. $s_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$, $s_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$ and $s_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$ correspond to reflections across the 3 angle bisectors. $s_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ and $s_5 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ correspond to rotations of $\frac{2}{3}\pi$ radians clockwise and counter-clockwise.

Example

Consider

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 1 & 7 & 6 & 9 & 4 & 2 & 5 & 8 & 10 \end{pmatrix} \in S_{10}$$

We note the following cycles

$$1 \rightarrow 3 \rightarrow 7 \rightarrow 2 \rightarrow 1$$

$$4 \rightarrow 6 \rightarrow 4$$

$$5 \rightarrow 9 \rightarrow 8 \rightarrow 5$$

$$10 \rightarrow 10$$

Thus, σ can be decomposed into one 4-cycle (1 3 7 2), one 2-cycle (4 6) and one 3-cycle (5 9 8) and one 1-cycle (10). Note that these cycles are pointwise disjoint and we have $\sigma = (1372)(46)(598)$.

Notation

$$(1372) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 1 & 7 & 4 & 5 & 6 & 2 & 8 & 9 & 10 \end{pmatrix}$$

$$(46) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 1 & 2 & 3 & 6 & 5 & 4 & 7 & 8 & 9 & 10 \end{pmatrix}$$

$$(598) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 1 & 2 & 3 & 4 & 9 & 6 & 7 & 5 & 8 & 10 \end{pmatrix}$$

$$(10) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{pmatrix}$$

Note

1. We do not usually write 1-cycles explicitly.
2. We can also write $\sigma = (46)(598)(1372) = (64)(985)(7213)$.

Theorem 1.3.5**Cycle Decomposition Theorem**

If $\sigma \in S_n$ with $\sigma \neq \varepsilon$, then σ is a product of (one or more) disjoint cycles of length at least 2. This factorization is unique up to the order of the factors.

Proof: See bonus.

□

Notation

Every permutation in S_n can be regarded as a permutation in S_{n+1} by fixing the number $(n+1)$. Thus, $S_1 \subseteq S_2 \subseteq \dots \subseteq S_n \subseteq S_{n+1} \subseteq \dots$

1.4. Cayley Tables

Definition 1.4.1

For a finite group G defining its operation by means of a table is sometimes convenient. Given $x, y \in G$, the product xy is the entry of the table in the row corresponding to x and the column corresponding to y . Such a table is a Cayley table.

Remark

By cancellation, the entries in each row or column of a Cayley table are all distinct.

Example

Consider $(\mathbb{Z}_2, +)$. Its Cayley table is

$\mathbb{Z}_2$	[0]	[1]
[0]	[0]	[1]
[1]	[1]	[0]

Example

Consider the group $\mathbb{Z}^* = \{1, -1\}$ under multiplication. Its Cayley table is

$\mathbb{Z}^*$	1	-1
1	1	-1
-1	-1	1

Note

If we replace 1 by [0] and -1 by [1], then the Cayley tables of $\mathbb{Z}^*$ and $\mathbb{Z}_2$ become the same. In this case, we say $\mathbb{Z}^*$ and $\mathbb{Z}_2$ are isomorphic, denoted by $\mathbb{Z}^* \cong \mathbb{Z}_2$.

Definition 1.4.2

For $n \in \mathbb{N}$, the cyclic group of order n is defined by $C_n = \{1, a, a^2, \dots, a^{n-1}\}$, with $a^n = 1$ and $1, a, \dots, a^{n-1}$ are distinct. The Cayley table of C_n is as follows.

C_n	1	a	a^2	...	a^{n-2}	a^{n-1}
1	1	a	a^2	...	a^{n-2}	a^{n-1}
a	a	a^2	a^3	...	a^{n-1}	1
a^2	a^2	a^3	a^4	...	1	a
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
a^{n-2}	a^{n-2}	a^{n-1}	1	...	a^{n-4}	a^{n-3}
a^{n-1}	a^{n-1}	1	a	...	a^{n-3}	a^{n-2}

Proposition 1.4.3

Let G be a group up to isomorphism. We have

1. If $|G| = 1$, then $G \cong \{1\}$.
2. If $|G| = 2$, then $G \cong C_2$.
3. If $|G| = 3$, then $G \cong C_3$.
4. If $|G| = 4$, then $G \cong C_4$ or $G \cong K_4 \cong C_2 \times C_2$, the Klein 4-group.

Proof:

1. If $|G| = 1$, then $G \cong \{1\}$.
2. If $|G| = 2$, then $G = \{1, g\}$ with $g \neq 1$. Then $g^2 = g$ or $g^2 = 1$. We note that if $g^2 = g$, by cancellation, we get $g = 1$, a contradiction. Therefore, $g^2 = 1$. Hence, the Cayley table of G is

G	1	g
1	1	g
g	g	1

3.

If $|G| = 3$, then $G = \{1, g, h\}$ with $g \neq 1, h \neq 1, g \neq h$. By cancellation, we have $gh \neq g, gh \neq h$. Thus, $gh = 1$. Similarly, we have $hg = 1$. Also, on the row for g , we have $g \cdot 1 = g, g \cdot h = 1$. Since all entries in this row are distinct, we have $g^2 = h$. Similarly, we have $h^2 = g$. Thus, we obtain the following Cayley table:

G	1	g	h
1	1	g	h
g	g	h	1
h	h	1	g

□

Exercise 1.4.4

Consider the symmetry group of a non-square rectangle. How is it related to K_4 ?

2. Subgroups

2.1. Subgroups

Definition 2.1.1

Let G be a group and $H \subseteq G$. If H itself is a group, then we say that H is a subgroup of G .

We note that since G is a group, for $h_1, h_2, h_3 \in H \subseteq G$, we have $h_1(h_2h_3) = (h_1h_2)h_3$. Thus, H is a subgroup of G if it satisfies the Subgroup Test.

Definition 2.1.2**The Subgroup Test**

Let G be a group and $H \subseteq G$. H is a subgroup if the following are true:

1. If $h_1, h_2 \in H$, then $h_1h_2 \in H$.
2. $1_H \in H$.
3. If $h_1 \in H$, then $h^{-1} \in H$.

Exercise 2.1.3

Prove that $1_H = 1_G$.

Proof: Let $H \subseteq G$ be a subgroup. Then $\forall a \in G, a \cdot 1_G = a = 1_G \cdot a$. But $H \subseteq G$, so $\forall a \in H, a \cdot 1_G = a = 1_G \cdot a$. Hence, $1_H = 1_G$. □

Example

Given a group G , then $\{1\}$ and G are subgroups of G .

Example

We have a chain of groups $(\mathbb{Z}, +) \subseteq (\mathbb{Q}, +) \subseteq (\mathbb{R}, +) \subseteq (\mathbb{C}, +)$.

Example

We recall the general linear group of order n over $\mathbb{R}$ is

$$GL_n(\mathbb{R}) = (GL_n(\mathbb{R}), \cdot) := \{M \in M_n(\mathbb{R}), \det(M) \neq 0\}$$

Define

$$SL_n(\mathbb{R}) = (SL_n(\mathbb{R}), \cdot) := \{M \in M_n(\mathbb{R}), \det(M) = 1\} \subseteq GL_n(\mathbb{R})$$

Note that the identity matrix $I \in SL_n(\mathbb{R})$. Let $A, B \in SL_n(\mathbb{R})$. Then $\det(AB) = \det(A)\det(B) = 1 \cdot 1 = 1$ and $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{1} = 1$. So $AB, A^{-1} \in SL_n(\mathbb{R})$. By the Subgroup Test, $SL_n(\mathbb{R})$ is a subgroup of $GL_n(\mathbb{R})$. We call $SL_n(\mathbb{R})$ the special linear group of order n over $\mathbb{R}$.

Definition 2.1.4

Given a group G , we define the center of G to be

$$Z(G) = \{z \in G : zg = gz \ \forall g \in G\}.$$

Note that $Z(G) = G$ iff G is abelian.

Proposition 2.1.5

$Z(G)$ is an abelian subgroup of G .

Proof:

Note that $1 \in Z(G)$. Let $y, z \in Z(G)$. Then for all $g \in G$, we have

$$(yz)g = y(zg) = y(gz) = (yg)z = (gy)z = g(yz)$$

Thus, $yz \in Z(G)$.

Also, for $z \in Z(G)$ and $g \in G$, we have

$$\begin{aligned} zg = gz &\Leftrightarrow z^{-1}(zg)z^{-1} = z^{-1}(gz)z^{-1} \\ &\Leftrightarrow (zz^{-1})gz^{-1} = z^{-1}g(zz^{-1}) \\ &\Leftrightarrow gz^{-1} = z^{-1}g \end{aligned}$$

Thus, $z^{-1} \in Z(G)$. By the Subgroup Test, $Z(G)$ is a subgroup of G . Also, by the definition of $Z(G)$, we see that it is abelian. $\square$

Proposition 2.1.6

Let H and K be subgroups of a group G . Then $H \cap K = \{g \in G : g \in H \wedge g \in K\}$ is a subgroup of G .

Proof: Let $f, g \in H \cap K$. Then $f, g \in H$, so $f \cdot g \in H$ since H is a group. Similarly, $f, g \in K$, so $f \cdot g \in K$ since K is a group. Therefore, $f \cdot g \in H \cap K$. Now, $1_G \in H \cap K$ since $1_H = 1_G$ by [Exercise 2.1.3](#) and $1_H \in H$, and since $1_K = 1_G$ by [Exercise 2.1.3](#) and $1_K \in K$. Finally, if $f \in H \cap K$, then $f \in H$, so $f_H^{-1} \in H$ since H is a group, and similarly, $f \in K$, so $f_K^{-1} \in K$. But $f_H^{-1} = f_K^{-1} = f_G^{-1} := f^{-1}$ since inverses are unique and since H, K are both subgroups of G . Therefore, $f^{-1} \in H \cap K$. Therefore, $H \cap K$ is a subgroup of G . $\square$

Proposition 2.1.7

Finite Subgroup Test

If H is a finite non-empty subset of a group G , then H is a subgroup of G iff H is closed under its operation.

Proof:

$\Rightarrow$ Trivial

$\Leftarrow$ For $H \neq \emptyset$, let $h \in H$. Since H is closed under its operation, we have $h, h^2, h^4, \dots$ are all in H . Since H is finite, these elements are not all distinct. Thus, $h^n = h^{n+m}$ for some $n, m \in \mathbb{N}$. By cancellation, $h^m = 1$ and thus $1 \in H$. Also, $1 = h^{m-1}h$ implies that $h^{-1} = h^{m-1}$ and thus $h^{-1} \in H$. By the subgroup test, H is subgroup of G . $\square$

2.2. Alternating Groups

Definition 2.2.1

A transposition $\sigma \in S_n$ is a cycle of length 2, ie: $(a \ b)$ with $a, b \in \{1, 2, \dots, n\}$ and $a \neq b$.

Example

Consider $(1 \ 2 \ 4 \ 5) \in S_5$. Also, the composition $(1 \ 2)(2 \ 4)(4 \ 5)$ can be computed as

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 5 & 4 \\ 1 & 4 & 3 & 5 & 2 \\ 2 & 4 & 3 & 5 & 1 \end{bmatrix}$$

Thus, we have $(1 \ 2 \ 4 \ 5) = (1 \ 2)(2 \ 4)(4 \ 5)$.

Also, we can show that $(1 \ 2 \ 4 \ 5) = (2 \ 3)(1 \ 2)(2 \ 5)(1 \ 3)(2 \ 4)$.

We see from this example that the factorization into transpositions are NOT unique. However, one can prove the following theorem (see Bonus 2).

Theorem 2.2.2

Parity Theorem

If a permutation σ has two factorizations $\sigma = \gamma_1 \gamma_2 \dots \gamma_r = \mu_1 \mu_2 \dots \mu_s$, where each γ_i and μ_j is a transposition, then $r \equiv s \pmod{2}$

Definition 2.2.3

A permutation σ is even (or odd) if it can be written as a product of an even (or odd) number of transpositions. By [Theorem 2.2.2](#), a permutation is either even or odd, but not both.

Theorem 2.2.4

For $n \geq 2$, let A_n denote the set of all even permutations in S_n . Then

1. $\varepsilon \in A_n$
2. If $\sigma, \tau \in A_n$, then $\sigma\tau \in A_n$ and $\sigma^{-1} \in A_n$
3. $|A_n| = \frac{1}{2}n!$

From (1) and (2), we see that A_n is a subgroup of S_n , called the alternating group of degree n .

Proof:

1. We can write $\varepsilon = (12)(12)$. Thus, ε is even.
2. If $\sigma, \tau \in A_n$, we can write $\sigma = \sigma_1\sigma_2\ldots\sigma_r$ and $\tau = \tau_1\tau_2\ldots\tau_s$, where σ_i, τ_i are transpositions and r, s are even integers. Then $\sigma\tau = \sigma_1\ldots\sigma_r\tau_1\ldots\tau_s$ is a product of $(r + s)$ transpositions and thus $\sigma\tau \in A_n$. Also, we note that since σ_i is a transposition, we have $\sigma_i^2 = \varepsilon$ and thus $\sigma_i^{-1} = \sigma_i$. It follows that $\sigma^{-1} = (\sigma_1\ldots\sigma_r)^{-1} = \sigma_r\sigma_{r-1}\ldots\sigma_1$, which is an even permutation.
3. Let O_n denote the set of odd permutations in S_n . Thus, $S_n = A_n \cup O_n$ and the parity then implies that $A_n \cap O_n = \emptyset$. Since $|S_n| = n!$, to prove $|A_n| = \frac{1}{2}n!$, it suffices to show that $|A_n| = |O_n|$. Let $\gamma = (12)$ and let $f : A_n \rightarrow O_n$ be defined by $f(\sigma) = \gamma\sigma$. Since σ is even, we have $\gamma\sigma$ is odd. Thus, the map is well-defined. Also, if we have $\gamma\sigma_1 = \gamma\sigma_2$, then by cancellation, we get $\sigma_1 = \sigma_2$. Thus f is injective.

Finally, if $\tau \in O_n$, then $\sigma = \gamma\tau \in A_n$ and $f(\sigma) = \gamma\sigma = \gamma(\gamma\tau) = \gamma^2\tau = \tau$. Thus, f is surjective. It follows that f is a bijection. Thus, $|A_n| = |O_n|$. It follows that $|A_n| = \frac{1}{2}n! = |O_n|$. □

2.3. Orders of Elements**Notation**

If G is a group and $g \in G$, we denote

$$\langle g \rangle = \{g^\beta : \beta \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, g^0, g^1, g^2, \dots\}$$

Note that $1 = g^0 \in \langle g \rangle$. Also, if $x = g^m, y = g^n \in \langle g \rangle$ with $m, n \in \mathbb{Z}$, then $xy = g^m g^n = g^{m+n} \in \langle g \rangle$ and $x^{-1} = g^{-m} \in \langle g \rangle$. Therefore, by [Definition 2.1.2](#), we have the following proposition:

Proposition 2.3.1

If G is a group and $g \in G$, then $\langle g \rangle$ is a subgroup of G .

Definition 2.3.2

Let G be a group and $g \in G$. We call $\langle g \rangle$ the cyclic subgroup of G generated by g . If $G = \langle g \rangle$ for some $g \in G$, then we say G is cyclic and g is a generator of G .

Example

Consider $(\mathbb{Z}, +)$. Note that for $k \in \mathbb{Z}$, we can write

$$k = k \cdot 1 := \underbrace{1 + 1 + \dots + 1}_{k \text{ times}}$$

Note that when $k = 0$, $k = 1_{\text{id}} = 0 \in \mathbb{Z}$. Thus, $\langle \mathbb{Z}, + \rangle = \langle 1 \rangle$. Similarly, $\langle \mathbb{Z}, + \rangle = \langle -1 \rangle$. We observe that for any $n \in \mathbb{Z}$ with $n \neq \pm 1$, there exist no $k \in \mathbb{Z}$ such that $k \cdot n = 1$. Thus, ± 1 are the only generators of $(\mathbb{Z}, +)$.

Note

Let G be a group and $g \in G$. Suppose that there exists $k \in \mathbb{Z}, k \neq 0$ such that

$$g^k = 1$$

We can assume that $k \geq 1$ because if $k \leq -1$, then $g^{-k} = 1$. So, by the well-ordering principle, there exists the “smallest” positive integer n such that $g^n = 1$.

Definition 2.3.3

Let G be a group and $g \in G$. If n is the smallest positive integer such that $g^n = 1$, then we say that the order of g is n , denoted by $o(g) = n$. If no such n exists, then we say g has **infinite order** and write $o(g) = \infty$.

Proposition 2.3.4

Let G be a group and $g \in G$ satisfying $o(g) = n \in \mathbb{N}$. For $k \in \mathbb{Z}$, we have

1. $g^k = 1$ iff $n \mid k$
2. $g^k = g^m$ iff $k \equiv m \pmod{n}$
3. $\langle g \rangle = \{1, g, g^2, \dots, g^{n-1}\}$, where $1, g, g^2, \dots, g^{n-1}$ are all distinct. In particular, we have $|\langle g \rangle| = o(g)$.

Proof:

(1)($\Leftarrow$) If $n \mid k$, then $k = nq$ for some $q \in \mathbb{Z}$. Thus, $g^k = g^{nq} = (g^n)^q = 1^q = 1$.

($\Rightarrow$) By the division algorithm, we can write $k = nq + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < n$.

Since $g^k = 1$ and $g^n = 1$, we have $g^r = g^{k-nq} = g^k \cdot (g^n)^{-q} = 1 \cdot 1^{-q} = 1$. Since $0 \leq r < n$ and $o(g) = n$, we have $r = 0$ and hence $n \mid k$.

(2) Note that $g^k = g^m$ iff $g^{k-m} = 1$. By (1), we have $n \mid (k - m)$, which is equivalent to $k \equiv m \pmod{n}$.

(3) It follows from (2) that $1, g, g^2, \dots, g^{n-1}$ are all distinct. Clearly, we have $\{1, g, g^2, \dots, g^{n-1}\} \subseteq \langle g \rangle$. To prove the other inclusion, let $g^k \in \langle g \rangle$ for some $k \in \mathbb{Z}$. Write $k = nq + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then

$g^k = (g^n)^q \cdot g^r = 1^q \cdot g^r = g^r \in \{1, g, \dots, g^{n-1}\}$. Thus, we have $\langle g \rangle = \{1, g, \dots, g^{n-1}\}$. $\square$

Proposition 2.3.5

Let G be a group and $g \in G$ satisfying $o(g) = \infty$. For $k \in \mathbb{Z}$, we have

1. $g^k = 1$ iff $k = 0$
2. $g^k = g^m$ iff $k = m$
3. $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, g^0 = 1, g^1, g^2, \dots\}$, where g^i are all distinct.

Proposition 2.3.6

Let G be a group and $g \in G$ with $o(g) = n \in \mathbb{N}$. If $d \in \mathbb{N}$, then $o(g^d) = \frac{n}{\gcd(n, d)}$, where $\gcd(n, d)$ is the gcd of n and d . In particular, if $d \mid n$, then $\gcd(n, d) = d$ and $o(g^d) = \frac{n}{d}$.

Proof: Let $n_1 = \frac{n}{\gcd(n, d)}$ and $d_1 = \frac{d}{\gcd(n, d)}$. By a result in MATH 135, we have $\gcd(n_1, d_1) = 1$. Note that

$$(g^d)^{n_1} = (g^d)^{\frac{n}{\gcd(n, d)}} = (g^n)^{\frac{d}{\gcd(n, d)}} = 1$$

since $o(g) = n$. Thus, it remains to show that n_1 is the smallest such positive integer.

Suppose $(g^d)^r = 1$ with $r \in \mathbb{N}$. Since $o(g) = n$, by [Proposition 2.3.4](#), we have $n \mid dr$. Thus, there exists $q \in \mathbb{Z}$ such that $dr = nq$. Dividing both sides by $\gcd(n, d)$, we get

$$d_1 r = \frac{d}{\gcd(n, d)} r = \frac{n}{\gcd(n, d)} q = n_1 q$$

Since $n_1 \mid d_1 r$ and $\gcd(n_1, d_1) = 1$, then by a result in MATH 135, we get $n_1 \mid r$, ie: $r = n_1 l$ for some $l \in \mathbb{Z}$. Since $r_1, n_1 \in \mathbb{N}$, it follows that $l \in \mathbb{N}$. Since $l \geq 1$, we get $r \geq n_1$. $\square$

2.4. Cyclic Groups

Remark

For a group G , if $G = \langle g \rangle$ for some $g \in G$, then G is a cyclic group. For $a, b \in G$, we have $a = g^m$ and $b = g^n$ for some $m, n \in \mathbb{Z}$ for some $m, n \in \mathbb{Z}$. We have $ab = g^m g^n = g^{m+n} = g^{n+m} = g^n g^m = ba$.

Proposition 2.4.1

Every cyclic group is abelian.

Remark

The converse of Proposition 2.4.1 is false. For example, the Klein 4 group $K_4 \cong C_2 \times C_2$ is abelian, but K_4 is not cyclic.

Proposition 2.4.2

Every subgroup of a cyclic group is cyclic.

Proof: Let $G = \langle g \rangle$ be cyclic and let H be a subgroup of G . If $H = \{1\}$, then $H = \langle 1 \rangle$ is cyclic.

If $H \neq \{1\}$, then there exists $g^k \in H$ with $k \in \mathbb{Z}$ and $k \neq 0$. Since H is a group, then we have $g^{-k} \in H$. Thus we can assume that $k \in \mathbb{N}$. Let m be the smallest positive integer such that $g^m \in H$.

Claim: $H = \langle g^m \rangle$

Proof of claim: Clearly, $\langle g^m \rangle \subseteq H$. For the other containment, let $g' \in H$. Then $g' = g^{mq+r}$ for some $q \in \mathbb{Z}$, $0 \leq r < m$, since $H \subseteq G = \langle g \rangle$ and by division algorithm. So $g' = (g^m)^q g^r$. Since H is a group and $g^m \in H$, then $(g^m)^q \in H$. So $g^r = (g^m)^{-q} g' \in H$. However, m is the smallest positive integer for which $g^r \in H$, so $r = 0$. Hence, $g' = (g^m)^q g^0 = (g^m)^q \cdot 1 = (g^m)^q \in \langle g^m \rangle$. So $H = \langle g^m \rangle$, as required. $\square$

So in either case, H is a cyclic subgroup of G , as required. $\square$

Proposition 2.4.3

Let G be a cyclic group with $o(g) = n$. Then $G = \langle g^k \rangle$ if and only if $\gcd(k, n) = 1$.

Proof: By Prop 2.3.6, $o(g^k) = \frac{n}{\gcd(n, k)} = n$ $\square$

Theorem 2.4.4

Fundamental Theorem of Finite Cyclic Groups

Let $G = \langle g \rangle$ be a cyclic group with $o(g) = n \in \mathbb{N}$.

1. If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $d \mid n$. It follows that $|H| \mid |G|$ ($|H|$ divides $|G|$).
2. Conversely, if $k \mid n$, then $\langle g^{\frac{n}{k}} \rangle$ is the unique subgroup of G of order k .

Proof:

1. By Proposition 2.4.2, H is cyclic. Write $H = \langle g^m \rangle$ for some $m \in \mathbb{N} \cup \{0\}$. Let $d = \gcd(m, n)$.

Claim: $H = \langle g^d \rangle$.

Proof of claim: Since $d \mid m$, we have $m = dk$ for some $k \in \mathbb{Z}$. Then

$g^m = g^{dk} = (g^d)^k \in \langle g^d \rangle$. Thus, $H = \langle g^m \rangle \subseteq \langle g^d \rangle$. To prove the other inclusion, since $d = \gcd(m, n)$, then by Bezout's Lemma, there exist $x, y \in \mathbb{Z}$ such that $d = mx + ny$. Then $g^d = g^{mx+ny} = (g^m)^x (g^n)^y = (g^m)^x \cdot 1^y = (g^m)^x \in \langle g^m \rangle$. Thus $\langle g^d \rangle \subseteq \langle g^m \rangle = H$. It follows that $H = \langle g^d \rangle$. Note that since $d = \gcd(m, n)$, we have $d \mid n$. By [Proposition 2.3.4](#) and [Proposition 2.3.6](#), we have

$$|H| = o(g^d) = \frac{n}{\gcd(n, d)} = \frac{n}{d}$$

Thus, $|H| \mid |G|$. □

2. By Prop 2.8, the cyclic subgroup $\langle g^{\frac{n}{k}} \rangle$ is of order

$$\frac{n}{\gcd(n, \frac{n}{k})} = \frac{n}{\frac{n}{k}} = k$$

To show uniqueness, let K be a subgroup of G , which is of order k with $k \mid n$. By (1), let $K = \langle g^d \rangle$ with $d \mid n$. Then by Prop 2.6 and Prop 2.8, we have

$$k = |K| = O(g^d) = \frac{n}{\gcd(n, d)} = \frac{n}{d}. \text{ It follows that } d = \frac{n}{k} \text{ and thus } K = \langle g^{\frac{n}{k}} \rangle. \quad \square$$

2.5. Non-Cyclic Groups

Definition 2.5.1

Let X be a non-empty subset of a group G , and let $\langle X \rangle = \{x_1^{k_1} x_2^{k_2} \dots x_m^{k_m} : x_i \in X, k_i \in \mathbb{Z}, m \geq 1\}$ denote the set of all products of powers of (not necessarily distinct) element of X .

Note

If $x_1^{k_1}, \dots, x_m^{k_m} \in \langle X \rangle$ and $y_1^{r_1} \dots y_n^{r_n} \in \langle X \rangle$, then $x_1^{k_1} \dots x_m^{k_m} y_1^{r_1} \dots y_n^{r_n} \in \langle X \rangle$. Also, $x_1^0 \in \langle X \rangle$ and $(x_1^{k_1} \dots x_m^{k_m})^{-1} = x_m^{-k_m} \dots x_1^{-k_1} \in \langle X \rangle$. Hence, $\langle X \rangle$ is a subgroup of G containing X , called the subgroup of G generated by X .

Example

The Klein 4-group $K_4 = \{1, a, b, c\}$ with $a^2 = b^2 = c^2 = 1$ and $ab = c$ (or $ac = b$ or $bc = a$). Thus, $K_4 = \langle a, b \mid a^2 = 1 = b^2 \wedge ab = ba \rangle$.

Example

The symmetric group of order 3, $S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$, where $\sigma^3 = \varepsilon = \tau^2$ and $\sigma\tau = \tau\sigma^2$ (one can take $\sigma = (1\ 2\ 3)$ and $\tau = (12)$). Thus,

$$S_3 = \langle \sigma, \tau \mid \sigma^3 = \varepsilon = \tau^2 \wedge \sigma\tau = \tau\sigma^2 \rangle$$

We can also replace σ, τ by $\sigma, \tau\sigma$ or $\sigma, \tau\sigma^2$, etc.

Definition 2.5.2

For $n \geq 2$, the dihedral group of order $2n$ is defined by

$$D_{2n} = \{1, a, \dots, a^{n-1}, b, ba, \dots, ba^{n-1}\}$$

where $a^n = 1 = b^2$ and $aba = b$. Thus,

$$D_{2n} = \langle a, b \mid a^n = 1 = b^2 \wedge aba = b \rangle$$

Note that when $n = 2$ or $n = 3$, we have $D_4 \cong K_4$ and $D_6 \cong S_3$.

Exercise 2.5.3

For $n \geq 3$, consider a regular n -gon and its group of symmetries. How does it relate to D_{2n} ?

3. Normal Subgroups

3.1. Homomorphisms and Isomorphisms

Definition 3.1.1

Let G and H be groups. A mapping $\alpha : G \rightarrow H$ is a homomorphism (HM) if

$$\alpha(a *_G b) = \alpha(a) *_H \alpha(b) \quad \forall a, b \in G$$

To simplify notation, we often write

$$\alpha(ab) = \alpha(a)\alpha(b) \quad \forall a, b \in G$$

Example

Consider the determinant map

$$\det = (GL_n(\mathbb{R}), \cdot) \rightarrow \mathbb{R}^*$$

given by

$$A \rightarrow \det(A)$$

Since $\det(AB) = \det(A) \det(B)$, the mapping $\det$ is a homomorphism.

Proposition 3.1.2

Let $\alpha : G \rightarrow H$ be a group HM. Then

1. $\alpha(1_G) = 1_H$
2. $\alpha(g^{-1}) = \alpha(g)^{-1} \quad \forall g \in G$
3. $\alpha(g^k) = \alpha(g)^k \quad \forall g \in G, \forall k \in \mathbb{Z}$

Proof:

1. $\alpha(1_G) = \alpha(1_G \cdot 1_G) = \alpha(1_G) \cdot \alpha(1_G)$, so by cancellation, $\alpha(1_G) = 1_H$.
2. $1_H = \alpha(1_G) = \alpha(g \cdot g^{-1}) = \alpha(g) \cdot \alpha(g^{-1})$. So $\alpha(g)^{-1} = \alpha(g^{-1})$.
- 3.

$$\alpha(g^k) = \alpha(\underbrace{g \dots g}_{k \text{ times}}) = \underbrace{\alpha(g) \dots \alpha(g)}_{k \text{ times}} = \alpha(g)^k$$

□

Definition 3.1.3

Let G and H be groups. Consider a mapping $\alpha : G \rightarrow H$. If α is a HM and α is bijective, we say α is an isomorphism (IM). In this case, we say G and H are isomorphic and denote it as $G \cong H$.

Proposition 3.1.4

We have

1. The identity map $G \rightarrow G$ is an IM.
2. If $\sigma : G \rightarrow H$ is an IM, then the inverse map $\sigma^{-1} : H \rightarrow G$ is also an IM.
3. If $\sigma : G \rightarrow H$ and $\tau : H \rightarrow K$ is an IM, then the composite map $\tau\sigma : G \rightarrow K$ is an IM.

Thus, we see that $\cong$ is an equivalence relation on the class of all groups.

Exercise 3.1.5

Proof of Proposition 3.1.4

1. The identity map is clearly bijective. Moreover, for any $g \in G$, $\text{id}(g \cdot g) = g \cdot g = \text{id}(g) \cdot \text{id}(g)$, so id is a homomorphism. Hence, id is an isomorphism.
2. Suppose σ is an isomorphism. Then σ is bijective. So σ^{-1} is bijective by a result from PMATH 351. Now, for $a, b \in H$, we have that

$$\begin{aligned}
 \sigma^{-1}(a \cdot b) &= \sigma^{-1}(\sigma(c) \cdot \sigma(d)) \quad (\text{for some } c, d \in G, \text{ since } \sigma \text{ is bijective}) \\
 &= \sigma^{-1}(\sigma(c \cdot d)) \quad (\text{since } \sigma \text{ is a homomorphism}) \\
 &= c \cdot d \\
 &= \sigma^{-1}(a) \cdot \sigma^{-1}(b) \quad (\text{since } a = \sigma(c) \Rightarrow c = \sigma^{-1}(a) \text{ and} \\
 &\quad b = \sigma(d) \Rightarrow d = \sigma^{-1}(b))
 \end{aligned}$$

Therefore, σ^{-1} is a homomorphism. Hence, σ^{-1} is an isomorphism.

3. Suppose σ and τ are isomorphisms. Then they are bijections, so $\tau\sigma$ is a bijection by a result from PMATH 351. Now, for $a, b \in G$, we have that

$$\begin{aligned}
 (\tau\sigma)(a \cdot b) &= \tau(\sigma(a \cdot b)) \\
 &= \tau(\sigma(a) \cdot \sigma(b)) \quad (\text{Since } \sigma \text{ is a homomorphism}) \\
 &= \tau(\sigma(a)) \cdot \tau(\sigma(b)) \quad (\text{Since } \tau \text{ is a homomorphism}) \\
 &= (\tau\sigma)(a) \cdot (\tau\sigma)(b)
 \end{aligned}$$

So $\tau\sigma$ is a homomorphism. Hence, $\tau\sigma$ is an isomorphism.

Example

Let $\mathbb{R}^+ = \{r \in \mathbb{R}, r > 0\}$.

Claim: $(\mathbb{R}, +) \cong (\mathbb{R}^+, \cdot)$

Proof: Define $\sigma : (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, \cdot)$ by $\sigma(r) = e^r$, where e is the exponential function. Note that the exponential map from $\mathbb{R}$ to $\mathbb{R}^+$ is bijective. Also, for $r, s \in \mathbb{R}$, we have $\sigma(r + s) = e^{r+s} = e^r \cdot e^s = \sigma(r) \cdot \sigma(s)$. So σ is a homomorphism. Thus, σ is isomorphism. □

Example

Claim: $(\mathbb{Q}, +)$ is not isomorphic to $(\mathbb{Q}^*, \cdot)$.

Proof: Suppose $\tau : (\mathbb{Q}, +) \rightarrow (\mathbb{Q}^*, \cdot)$ is an isomorphism. Thus, τ is surjective. So there exists $q \in \mathbb{Q}$ such that $\tau(q) = 2$. Write $\tau(\frac{q}{2}) = a \in \mathbb{Q}^* \subseteq \mathbb{Q}$. Since τ is a HM, we have

$$a^2 = \tau\left(\frac{q}{2}\right)\tau\left(\frac{q}{2}\right) = \tau\left(\frac{q}{2} + \frac{q}{2}\right) = \tau(q) = 2$$

which contradicts the fact that $a \in \mathbb{Q}$. Thus, such a τ does not exist and hence $(\mathbb{Q}, +) \not\cong (\mathbb{Q}^*, \cdot)$. □

3.2. Cosets and Lagrange's Theorem**Definition 3.2.1**

Let H be a subgroup of a group G . If $a \in G$, we define $Ha = \{ha \mid h \in H\}$ to be the right coset of H generated by a . Similarly, we define $aH = \{ah \mid h \in H\}$ to be the left coset of H generated by a . Since $1 \in H$, we have $H1 = H = 1H$ and $a \in Ha$ and $a \in aH$.

Note that in general, Ha and aH are not subgroups of G and $aH \neq Ha$. However, if G is abelian, then $Ha = aH$.

Example

Let $K_4 = \{1, a, b, ab\}$ with $a^2 = 1 = b^2$ and $ab = ba$. Let $H = \{1, a\}$, which is a subgroup of K_4 . Note that since K_4 is abelian, we have $gH = Hg \forall g \in K_4$. Then the right or left cosets of H are

$$H1 = \{1, a\} = 1H$$

and

$$Hb = \{b, ab\} = Hab$$

Thus, there are exactly two cosets of H in K_4 .

Example

Let $S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$ with $\sigma^3 = \varepsilon = \tau^2$ and $\sigma\tau\sigma = \tau$. Let $H = \{\varepsilon, \tau\}$ which is a subgroup of S_3 . Since $\sigma\tau = \tau\sigma^{-1} = \tau\sigma^2$, the right cosets of H are

$$H\varepsilon = \{\varepsilon, \tau\} = H\tau$$

$$H\sigma = \{\sigma, \tau\sigma\} = H\tau\sigma$$

$$H\sigma^2 = \{\sigma^2, \tau\sigma^2\} = H\tau\sigma^2$$

Also, the left cosets of H are

$$\varepsilon H = \{\varepsilon, \tau\} = \tau H$$

$$\sigma H = \{\varepsilon, \tau\sigma^2\} = \tau\sigma^2 H$$

$$\sigma^2 H = \{\sigma^2, \tau\sigma, \tau\sigma H\}$$

Note that $H\sigma \neq \sigma H$ and $H\sigma^2 \neq \sigma^2 H$

Proposition 3.2.2

Let H be a subgroup of a group G and let $a, b \in G$. Then

1. $Ha = Hb$ iff $ab^{-1} \in H$. In particular, we have $Ha = H$ iff $a \in H$ (taking $b = 1$).
2. If $a \in Hb$, then $Ha = Hb$.
3. Either $Ha = Hb$ or $Ha \cap Hb = \emptyset$. Thus, the distinct right cosets of H form a partition of G .

Proof:

1. ($\implies$) If $Ha = Hb$, then $a = 1a \in Ha = Hb$. Thus, $a = hb$ for some $h \in H$ and we have $ab^{-1} = h \in H$.
- ($\impliedby$) Suppose $ab^{-1} \in H$. Then for all $h \in H$,

$$ha = (ha)(b^{-1}b) = h(ab^{-1})b \in Hb$$

Thus $Ha \subseteq Hb$. Note that if $ab^{-1} \in H$, since H is a subgroup, then

$(ab^{-1})^{-1} = ba^{-1} \in H$. Thus for all $h \in H$, $hb = h(ba^{-1})a \in Ha$. Thus $Hb \subseteq Ha$. It follows that $Ha = Hb$.

2. If $a \in Hb$, then $ab^{-1} \in H$. Thus by (1), we have $Ha = Hb$. Two cases:
 - a. If $Ha \cap Hb = \emptyset$, then we are done.
 - b. If $Ha \cap Hb \neq \emptyset$, then there exists $x \in Ha \cap Hb$. Since $x \in Ha$, by (2), we have $Hx = Ha$. Since $x \in Hb$, by (2), we have $Hb = Hx$. Thus $Ha = Hx = Hb$.

□

Remark

The analog of [Proposition 3.2.2](#) also holds for left cosets. For (1), we have $aH = bH$ iff $b^{-1}a \in H$.

Exercise 3.2.3

Let G be a group and $H \subseteq G$ be a subset of G . For $a, b \in G$, do we still have $Ha = Hb$, or $Ha \cap Hb = \emptyset$ if H is not a subgroup of G ?

Definition 3.2.4

By [Proposition 3.2.2](#), we see that G can be written as a disjoint union of right cosets of H . We define the *index* $[G = H]$ to be the number of disjoint right (or left) cosets of H in G (note that $[G = H]$ could be infinite).

Theorem 3.2.5**Lagrange's Theorem**

Let H be a subgroup of a finite group G . We have $|H| \mid |G|$ and $[G = H] = \frac{|G|}{|H|}$.

Proof: Write $k = [G = H]$ and let $Ha_1, Ha_2, \dots, Ha_k$ be the distinct right cosets of H in G . By [Proposition 3.2.2](#), we have $G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k$ is a disjoint union. Since $|Ha_i| = |H|$ for each i , we have $|G| = |Ha_1| + |Ha_2| + \dots + |Ha_k| = k |H|$. It follows that $|H| \mid |G|$ and

$$[G = H] = k = |G| \mid |H|$$

□

Corollary 3.2.6

1. If G is a finite group and $g \in G$, then $o(g) \mid |G|$.
2. If G is a finite group with $|G| = n$, then for all $g \in G$, we have $g^n = 1$.

Proof:

1. Take $H = \langle g \rangle$ in [Theorem 3.2.5](#). Note that $|H| = o(g)$.
2. Let $o(g) = m$. Then by (1), we have $m \mid n$. Thus $g^n = (g^m)^{\frac{n}{m}} = 1^{\frac{n}{m}} = 1$.

□

Example

For $n \in \mathbb{N}$ with $n \geq 2$, let $\mathbb{Z}_n^*$ be the set of (multiplicatively) invertible elements in $\mathbb{Z}_n$. Let the Euler's φ function, $\varphi(n)$, denote the order of $\mathbb{Z}_n^*$, ie:

$\varphi(n) = \#\{[k] \in \mathbb{Z}_n : k \in \{0, 1, \dots, n-1\} \wedge \gcd(k, n) = 1\}$. As a direct consequence of [Corollary 3.2.6](#), we see that if $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

This is Euler's Theorem. If $n = p$, a prime number, then Euler's theorem implies $a^{p-1} \equiv 1 \pmod{p}$, which is Fermat's Little Theorem from MATH 135.

Recall

If $|G| = 2$, $G \cong C_2$ and $|G| = 3$, $G \cong C_3$.

Corollary 3.2.7

If G is a group with $|G| = p$, a prime, then $G \cong C_p$, the cyclic group of order p .

Proof: Let $g \in G$ with $g \neq 1$. Then by [Corollary 3.2.6](#), we have $o(g) \mid p$. Since $g \neq 1$ and p is prime, we have $o(g) = p$. By Prop 2.6 (TODO), we have $|\langle g \rangle| = o(g) = p$. It follows that $G = \langle g \rangle \cong C_p$. $\square$

Corollary 3.2.8

Let H and K be finite subgroups of a group G . If $\gcd(|H|, |K|) = 1$, then $H \cap K = \{1\}$.

Proof: By [Proposition 2.1.6](#), $H \cap K$ is a subgroup of H and K . By [Theorem 3.2.5](#), we have $|H \cap K| \mid |H|$ and $|H \cap K| \mid |K|$. It follows that $|H \cap K| \mid \gcd(|H|, |K|)$, ie: $|H \cap K| \mid 1$. Thus $H \cap K = \{1\}$. $\square$

3.3. Normal Subgroups**Definition 3.3.1**

Let H be a subgroup of a group G . If $gH = Hg$ for all $g \in G$, we say H is *normal*, denoted by $H \triangleleft G$.

Example

We have $\{1\} \triangleleft G$ and $G \triangleleft G$.

Example

The center $Z(G)$ of G , $Z(G) = \{z \in G, zg = gz \ \forall g \in G\}$ is an abelian subgroup of G . By its definition, $Z(G) \triangleleft G$. Thus, every subgroup of $Z(G)$ is normal in G .

Example

If G is an abelian group, then every subgroup of G is normal in G . The converse is false (see the quaternion group Q_8 in A3).

Proposition 3.3.2**Normality Test**

Let H be a subgroup of a group G . The following are equivalent:

1. $H \triangleleft G$
2. $gHg^{-1} \subseteq H$ for all $g \in G$. We call gHg^{-1} a *conjugate* of H .
3. $gHg^{-1} = H$ for all $g \in G$ (thus $H \triangleleft G$ iff H is the only conjugate of H)

Proof:

- (1) $\Rightarrow$ (2) Let $ghg^{-1} \in gHg^{-1}$ for some $h \in H$. Then by (1), $gh \in gH = Hg$, say $gh = h_1g$ for some $h_1 \in H$. Then $ghg^{-1} = h_1gg^{-1} = h \in H$.
- (2) $\Rightarrow$ (3) If $g \in G$, then by (2), $gHg^{-1} \subseteq H$. Taking g^{-1} in place of g in (2), we get $g^{-1}Hg \subseteq H$. This implies that $H \subseteq gHg^{-1}$. Thus $H = gHg^{-1}$.
- (3) $\Rightarrow$ (1) If $gHg^{-1} = H$, then $gH = Hg$. □

Example

Let $G = GL_n(\mathbb{R})$ and $H = SL_n(\mathbb{R})$. For $A \in G$ and $B \in H$, we have

$$\det(ABA^{-1}) = \det(A)\det(B)\det(A^{-1}) = \det(A) \cdot 1 \cdot \frac{1}{\det(A)} = 1$$

Thus $ABA^{-1} \in H$ and it follows that $AHA^{-1} \subseteq H \forall A \in G$. By the normality test, $H \triangleleft G$.

Proposition 3.3.3

If H is a subgroup of a group G with $[G : H] = 2$, then $H \triangleleft G$.

Proof: Let $g \in G$. If $g \in H$, then $Hg = H = gH$. If $g \notin H$, since $[G : H] = 2$, then $G = H \cup Hg$, a disjoint union. Then $Hg = G \setminus H$. Similarly, $gH = G \setminus H$. Thus $gH = Hg$ for all $g \in G$, ie: $H \triangleleft G$. □

Example

Let A_n be the alternating group contained in S_n . Since $[S_n : A_n] = 2$, by Prop 3.9, we have $A_n \triangleleft S_n$.

Example

Let $D_{2n} = \langle a, b \mid a^n = 1 = b^2 \wedge aba = b \rangle$ be the dihedral group of order $2n$. Since $[D_{2n} : \langle a \rangle] = 2$, by Prop 3.9, $\langle a \rangle \triangleleft D_{2n}$.

Recall

Let H and K be subgroups of a group G . The intersection $H \cap K$ be the largest subgroup of G contained in both H and K .

Note

What is the smallest subgroup containing H and K ?

$H \cup K$ is the smallest subset containing H and K , but $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$ by A2.

A more useful subset to consider is the *product* HK of H and K defined as follows: $HK = \{hk \mid h \in H, k \in K\}$. But HK is not always a subgroup of G .

Lemma 3.3.4

Let H and K be subgroups of a group G . The following are equivalent:

1. HK is a subgroup of G .
2. $HK = KH$.
3. KH is a subgroup of G .

Proof: We will prove $(1) \Leftrightarrow (2)$. Then $(2) \Leftrightarrow (3)$ follows by interchanging H and K .

$(2) \Rightarrow (1)$ We have $1 = 1 \cdot 1 \in HK$. Also, if $hk \in HK$, then

$(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$. Also, for $h, k, h_1, k_1 \in HK$, we have

$kh_1 \in KH = HK$, say $kh_1 = h_2k_2$. It follows that

$(hk)(h_1k_1) = h(kh_1)k_1 = h(h_2k_2)k_1 = (hh_2)(k_2k_1) \in HK$. By the subgroup test, HK is a subgroup of G .

$(1) \Rightarrow (2)$ Let $kh \in KH$ with $k \in K$ and $h \in H$. Since H and K are subgroups of G , we have $h^{-1} \in H$ and $k^{-1} \in K$. Since HK is a subgroup of G , we have

$kh = (h^{-1}k^{-1})^{-1} \in HK$. Thus $KH \subseteq HK$. Similarly, one can show that $HK \subseteq KH$. $\square$

Proposition 3.3.5

Let H and K be subgroups of a group G .

1. If $H \triangleleft G$ or $K \triangleleft G$, then $KH = HK$ is a subgroup of G .
2. If $H \triangleleft G$ and $K \triangleleft G$, then $HK \triangleleft G$.

Proof: (1) Suppose $H \triangleleft G$. Then

$$HK = \bigcup_{k \in K} Hk = \bigcup_{k \in K} kH = KH$$

By Lemma 3.3.4, $HK = KH$ is a subgroup of G .

(2) If $g \in G$ and $hk \in HK$, since $H \triangleleft G$ and $K \triangleleft G$, we have

$g^{-1}(hk)g = (g^{-1}hg)(g^{-1}kg) \in HK$. Thus $g^{-1}HKg \subseteq HK$ and $HK \triangleleft G$. $\square$

Definition 3.3.6

Let H be a subgroup of a group G . The *normalizer* of H , denoted by $N_G(H)$ is defined to be $N_G(H) = \{g \in G : gH = Hg\}$. We see that $H \triangleleft G$ if and only if $N_G(H) = G$.

Note

In the proof of [Proposition 3.3.5](#), we do not need the full assumption that $H \triangleleft G$. We only need $kH = Hk$ for all $k \in K$, ie: $k \in N_G(H)$. Thus, we have the following corollary:

Corollary 3.3.7

Let H and K be subgroups of a group G . If $K \subseteq N_G(H)$ (or $H \subseteq N_G(K)$), then $HK = KH$ is a subgroup of G .

Theorem 3.3.8

If $H \triangleleft G$ and $K \triangleleft G$ satisfy $H \cap K = \{1\}$, then $HK \cong H \times K$.

Proof: Claim: If $H \triangleleft G$ and $K \triangleleft G$ satisfy $H \cap K = \{1\}$, then $hk = kh$ for all $h \in H$ and $k \in K$.

Proof of of claim 1: Consider $x = hk(kh)^{-1} = hkh^{-1}k^{-1}$. Note that $kh^{-1}k^{-1} \in kHk^{-1} = H$ (since $H \triangleleft G$). Thus $x = h(kh^{-1}k^{-1}) \in H$. Similarly, since $hkh^{-1} \in hKh^{-1} = K$, we have $x = (hkh^{-1})k^{-1} \in K$. Since $x \in H \cap K = \{1\}$, we have $hkh^{-1}k^{-1} = 1$, ie: $hk = kh$. Since $H \triangleleft G$, by [Proposition 3.3.5](#), we have that HK is a subgroup of G . Define $\sigma : H \times K \rightarrow HK$ by $\sigma((h, k)) = hk$.

Claim: Let $(h, k) \in H \times K$. By Claim 1, we have $h_1k = kh_1$. Thus, $\sigma((h, k) \cdot (h_1, k_1)) = \sigma((hh_1, kk_1)) = hh_1kk_1 = hkh_1k_1 = \sigma((h, k))\sigma((h_1, k_1))$. Thus σ is a homomorphism. Note that by the definition of HK , σ is surjective. Also, if $\sigma((h, k)) = \sigma((h_1, k_1))$, we have $hk = h_1k_1$. Thus $h_1^{-1}h = k_1k^{-1} \in H \cap K = \{1\}$. Thus $h_1^{-1}h = 1 = k_1k^{-1}$, ie: $h_1 = h$ and $k_1 = k$. Thus σ is injective. Thus, σ is an isomorphism and we have $HK \cong H \times K$. □

Corollary 3.3.9

Let G be a finite group and let H and K be normal subgroups such that $H \cap K = \{1\}$ and $|H| |K| = |G|$. Then $G \cong H \times K$.

Example

Let $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. Let G be a cyclic group of order mn . Write $G = \langle a \rangle$ with $o(a) = mn$. Let $H = \langle a^n \rangle$ and $K = \langle a^m \rangle$. Thus $|H| = o(a^n) = m$ and $|K| = o(a^m) = n$. It follows that $|H| \cdot |K| = mn = |G|$. Since $\gcd(m, n) = 1$. By corollary 3.2.8 (TODO), $H \cap K = \{1\}$. Also, since G is cyclic and thus abelian, we have $H \triangleleft G$ and $K \triangleleft G$. Thus, by [Corollary 3.3.9](#), we have $G \cong H \times K$, ie: $C_{mn} \cong C_m \times C_n$. Hence to consider finite cyclic group, it suffices to consider cyclic groups of prime power order.

4. Isomorphism Theorems

4.1. Quotient Groups

Definition 4.1.1

Let K be a subgroup of a group G . Consider the set of right cosets of K , ie: $\{Ka : a \in G\}$. To make it a group, a natural way is to define

$$Ka \cdot Kb = K \cdot (ab) \quad \forall a, b \in G \quad (*)$$

Note that we could have $Ka = Ka_1$ and $Kb = Kb_1$ with $a \neq a_1$ and $b \neq b_1$. Thus, in order for $(*)$ to make sense, a necessary condition is

$$Ka = Ka \wedge Kb = Kb_1 \Rightarrow Kab = Ka_1b_1$$

In this case, we say that the multiplication $KaKb = Kab$ is *well-defined*.

Lemma 4.1.2

Let K be a subgroup of a group G . The following are equivalent:

1. $K \triangleleft G$
2. For $a, b \in G$, the multiplication $KaKb = Kab$ is well-defined.

Proof: (1) $\Rightarrow$ (2) Let $Ka = Ka_1$ and $Kb = Kb_1$. Thus $aa_1^{-1} \in K$ and $bb^{-1} \in K$. To get $Kab = Ka_1b_1$, we need $ab(a_1b_1)^{-1} \in K$. Note that since $K \triangleleft G$, we have $aKa^{-1} = K$. Thus $ab(a_1b_1)^{-1} = abb_1^{-1}a_1^{-1} = (abb_1^{-1}a^{-1})(aa_1^{-1}) \in K$. Thus $Kab = Ka_1b_1$. (2) $\Rightarrow$ (1) If $a \in G$, to show $K \triangleleft G$, we need $aka^{-1} \in K$ for all $k \in K$. Since $Ka = Ka$ and $Kk = K1$, by (2), we have $Kak = Ka1$, ie: $Kak = Ka$. It follows that $aka^{-1} \in K$. Thus $K \triangleleft G$. □